

Toward a Standard Benchmark Format and Suite for Floating-Point Analysis



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Incredible progress...

Automatic Verification

Fluctuat [SAS'13]

Rosa [POPL'14]

FPTaylor [FM'15]

Optimization

STOKE [PLDI'14]

Improvement

Salsa [FMICS'15]

Herbie [PLDI'15]

Mechanized Proofs

Wave equation [ITP'10]

Rounding error [NSV'16]

Rapid improvement in hard problems!

Incredible progress...

Auto
Fluc
Rosa
FPTa

We want our community
to keep progressing!

Next
???

Rapid improvement in hard problems!

Incredible progress

We want our community
to keep progressing!

Auto
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Rosa
FPTa

Next
???

Salsa
Herbie

As community grows,
growing pains appear

Rapid improvement in hard problems!

Growing pains

Similar growing pains in
compilers, HPC, SAT, SMT, ...
communities

Composition

Rosa → Salsa

~~Rosa: def example(x: Double): Double = ...~~
 ~~$\sqrt{x+1} - \sqrt{x}$~~
Salsa: double example(double x) { ... }

Evaluation

Fluctuat FPTaylor

Fluctuat: Poly, Inv, F1a, F1b, idem, ...
FPTaylor: sine, sqrt, verhulst, ...

Standardization

Herbie STOKE

Herbie: ulp(NaN, Inf) = UINT_MAX
STOKE: ulp(NaN, Inf) < UINT_MAX



FPBench

FPBench is **community infrastructure**
for **cooperation** and **comparison**
in the FP community.

<http://fpbench.org>



FPBench

β

Common format

Benchmark suite

Named measures



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$$\sqrt{x + 1} - \sqrt{x}$$

Arguments

(FPCore (x))

(- (sqrt (+ x 1)) (sqrt x)))

S-expression syntax

$$\sqrt{x + 1} - \sqrt{x}$$

```
(FPCore (x)
  :name "Sqrt Difference"
  :cite (hamming-87)
  :pre (> x 0)
  (- (sqrt (+ x 1)) (sqrt x)))
```

Metadata

Preconditions

```

(FPCore (x0)
  :name "Sine Newton"
  :cite (darulova-kuncak-2014)
  :pre (< (abs x0) 1)
  (while (< i 10)
    ([i 0 (+ i 1)]
     [x x0
      (+ (+ (- x (/ (pow x 3) 6.0))
                (/ (pow x 5) 120.0))
          (/ (pow x 7) 5040.0))
      (+ (+ (- 1.0 (/ (* x x) 2.0))
                (/ (pow x 4) 24.0))
          (/ (pow x 6) 720.0)))]))
    x))

```

Loops

Common functions

FPCore common format

Simple to use

S-expression syntax
Purely functional
No control flow analysis

Expressive

All C, Fortran functions
Loops, conditionals
Tools support parts

Extensible

Metadata properties
Tool-specific metadata
Input or output format



Generate from
higher-level,
imperative
FPImp lang.



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Named measures

Simple to implement

Covers all existing uses

Simple to extend, specialize



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FPBench benchmark suite

72 total benchmarks

Drawn from existing papers

Annotated with source, ranges, description, citation

FPBench benchmark suite

Existing programs

FPTaylor	29
Herbie	28
Rosa	6
Salsa	9

Rich features

Arith	72
Expt	16
Trig	11
Loop	12
Branch	3

Diverse domains

Textbook	59
Math Alg	6
Emb Sys	4
Sci Comp	3



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Common format

Benchmark suite

Named measures

Simple to implement

From existing projects

Covers all existing uses

Cover many domains

Simple to extend, specialize

Grows over time



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From existing projects

Cover many domains

Grows over time

FPBench measures

Formal definitions of accuracy measures

Described along 5 axes

Standard measures so tools agree

FPBench axes of measurement

Scaling vs. non-scaling

Absolute, relative, ULPs, bits, ...

Forward vs. backward

Fixed input error vs fixed output error

Maximum vs. average



Sound vs. statistical

Formal guarantees vs mathematical accuracy

Improvement





FPBench

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Benchmark suite

From existing projects

Cover many domains

Grows over time

Named measures

Terms for measuring error

Standard across tools

Flexible but rigorous



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